SET 8 – MATHEMATICS – IDENTITIES & SUMS

Question Paper | Total Marks: 15

Q1. Cube Identity (2 Marks)

Evaluate 101^3 using an identity.

Q2. Sum of Squares of Two Numbers (2 Marks)

If $a + b = 5$ and $ab = 6$, find $a^2 + b^2$.

Q3. Sum of Cubes of Two Numbers (3 Marks)

If $a + b = 5$ and $ab = 6$, find $a^3 + b^3$.

Q4. Sum of Cubes of Even Numbers (3 Marks)

Find $2^3 + 4^3 + 6^3 + \ldots + 16^3$.

Q5. Proof by Induction – Sum of Squares (5 Marks)

Prove that $1^2 + 2^2 + \ldots + n^2 = n(n + 1)(2n + 1)/6$.